

# CYBER SECURITY — MINOR PROJECT REPORT

Hands-on Implementation: Kali Linux OS, Network Spoofing & User Management

**Student Name:** Dharmendra Kurre  
**Course:** Cyber Security & Ethical Hacking

**Submission Deadline:** 20th September 2026  
**Platform:** VMware Workstation

## 1. INTRODUCTION

**About Kali Linux:** Kali Linux is an advanced, Debian-derived Linux distribution developed and maintained by Offensive Security. It is specifically crafted for penetration testing, security auditing, vulnerability assessment, and digital forensics. Kali Linux includes hundreds of pre-installed cybersecurity tools such as Nmap, Wireshark, Metasploit, Aircrack-ng, and John the Ripper.

**Importance in Cybersecurity:** Kali Linux serves as the standard operating system for both Red Team (offensive) and Blue Team (defensive) practitioners. Its pre-configured security packages, custom kernel supporting wireless packet injection, and isolated command-line environment make it essential for testing security posture and simulating cyber attacks safely.

## 2. OBJECTIVES

- Set up and configure Kali Linux within an isolated virtualized environment (VMware Workstation).
- Validate installation and verify operating system details via Linux terminal.
- Perform repository synchronization and package upgrades to ensure software security patching.
- Execute Media Access Control (MAC) address spoofing using macchanger for identity cloaking and network privacy.
- Implement role-based user management by provisioning standard users and delegating administrative sudo privileges.

## 3. ENVIRONMENT SETUP & PREREQUISITES

| Component               | Specification / Detail                                               |
|-------------------------|----------------------------------------------------------------------|
| Host Operating System   | Microsoft Windows 11 (64-bit)                                        |
| Virtualization Software | VMware Workstation (kali-linux-2026.2-vmware-amd64)                  |
| Target Virtual OS       | Kali GNU/Linux Rolling (Kernel 64-bit)                               |
| Resource Allocation     | Memory: 4 GB RAM   Processors: 2 Cores   Storage: 25 GB Virtual Disk |
| Networking Mode         | NAT / Bridged (Adapter: eth0)                                        |

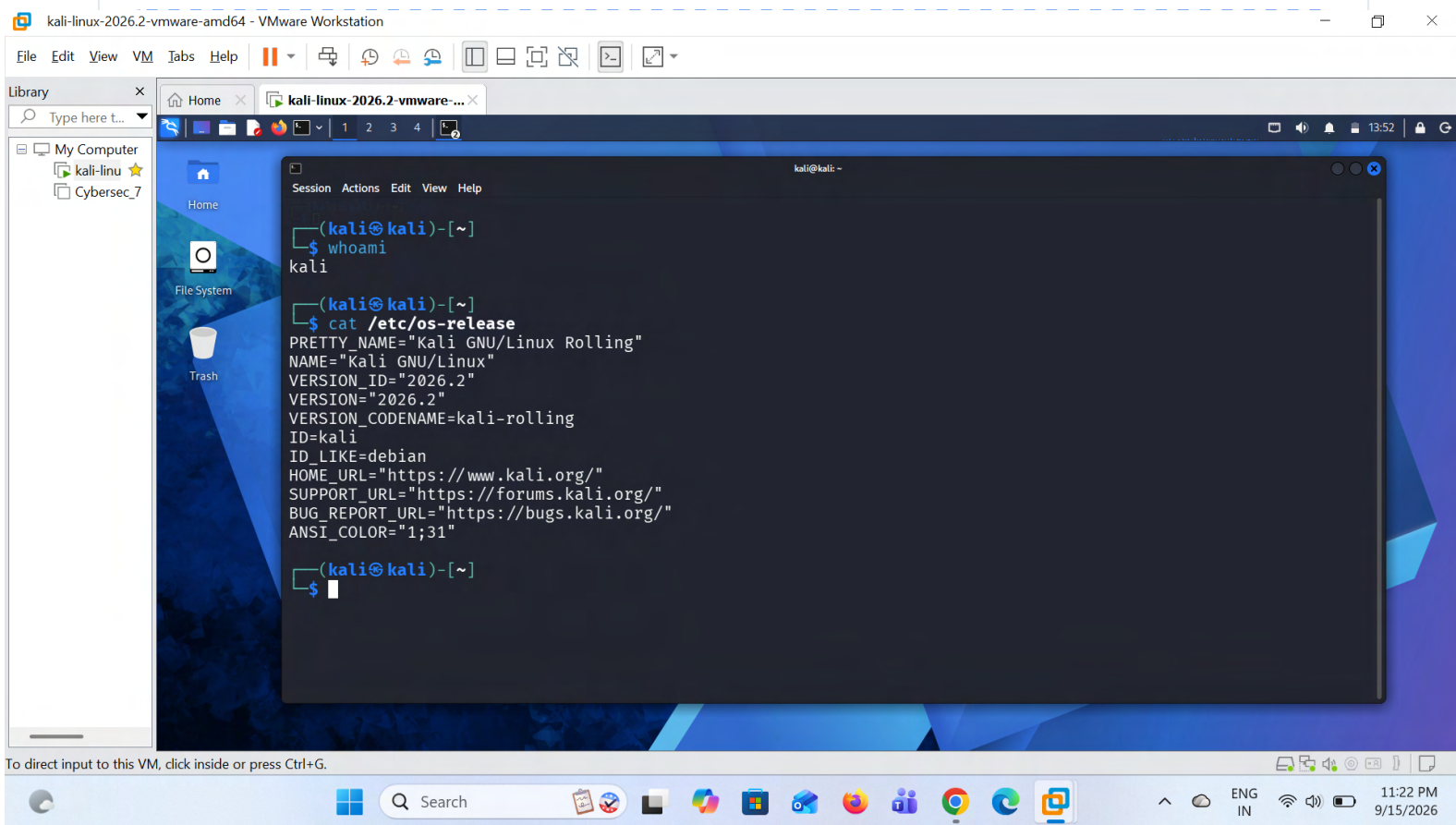
## 4. PRACTICAL TASKS & VERIFICATION

### Task 1: System Installation and First Login

**Cybersecurity Relevance:** Deploying Kali Linux in a virtual machine ensures complete network isolation, enabling practitioners to test exploits and monitor hostile network traffic without risking the host environment.

#### Commands Executed:

```
$ whoami
$ cat /etc/os-release
```



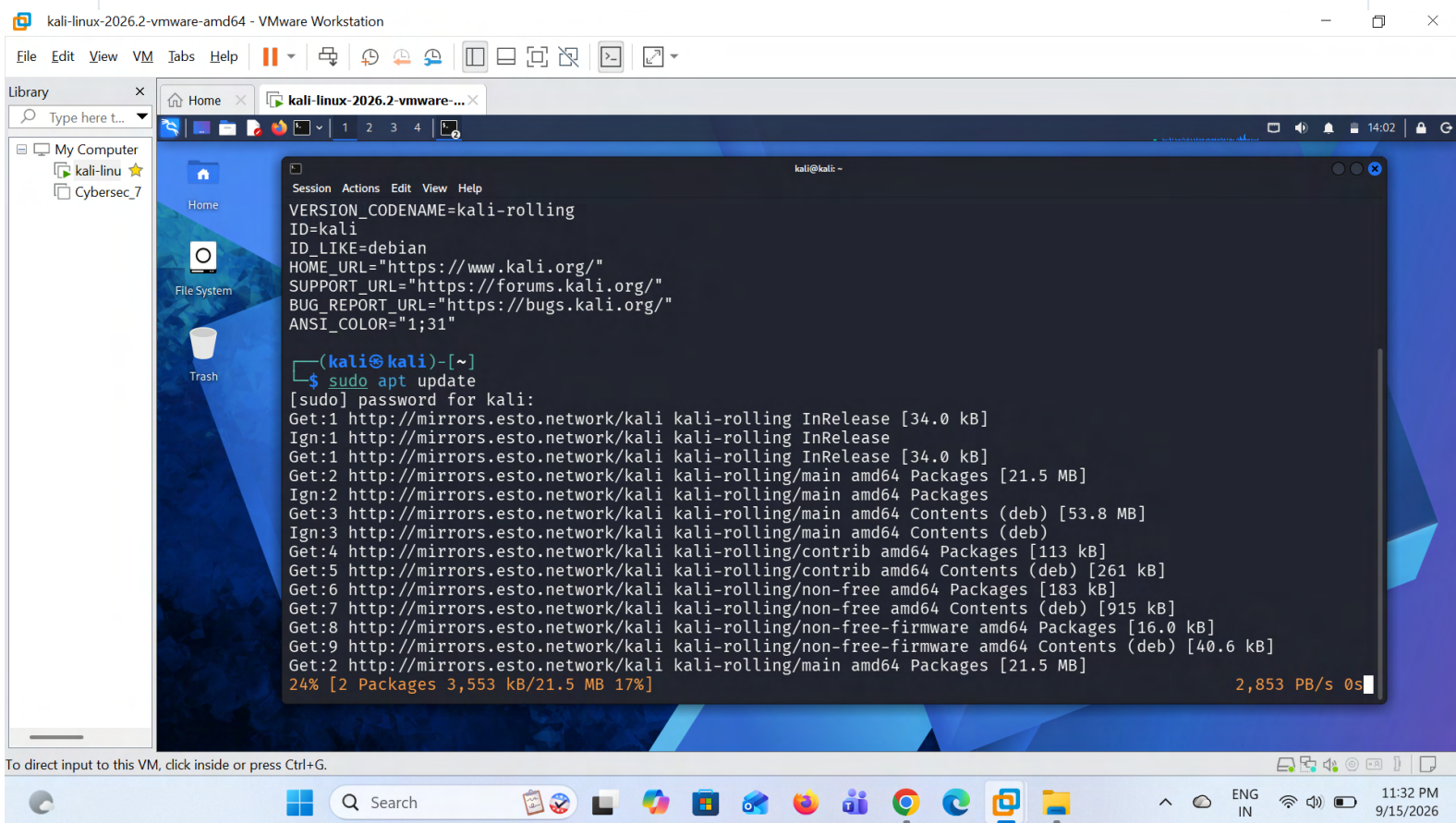
## PRACTICAL TASKS & VERIFICATION (CONTINUED)

### Task 2: System Repository Update and Upgrade

**Cybersecurity Relevance:** Maintaining up-to-date repositories is critical. Unpatched kernels or outdated security utilities can introduce vulnerabilities into testing infrastructure. Regular updates patch known CVEs and update signature databases.

#### Commands Executed:

```
$ sudo apt update
$ sudo apt upgrade -y
```



The screenshot shows a VMware Workstation window titled "kali-linux-2026.2-vmware-amd64 - VMware Workstation". Inside the VM, a terminal window is open, displaying the output of the command `sudo apt update`. The terminal output shows the system is using the "kali-rolling" repository and lists various packages being updated, including the kernel and security utilities. The progress bar at the bottom of the terminal indicates that 24% of the packages (2 out of 3,553 kB) have been updated, with a total size of 21.5 MB. The terminal also shows the system's configuration, including the home URL, support URL, and bug report URL. The VMware interface includes a menu bar (File, Edit, View, VM, Tabs, Help) and a toolbar with icons for various VM functions. The desktop background of the VM is a blue and white geometric pattern. The VMware window's title bar shows the current time as 14:02 and the date as 9/15/2026.

```
Session Actions Edit View Help
VERSION_CODENAME=kali-rolling
ID=kali
ID_LIKE=debian
HOME_URL="https://www.kali.org/"
SUPPORT_URL="https://forums.kali.org/"
BUG_REPORT_URL="https://bugs.kali.org/"
ANSI_COLOR="1;31"

(kali@kali)-[~]
$ sudo apt update
[sudo] password for kali:
Get:1 http://mirrors.esto.network/kali kali-rolling InRelease [34.0 kB]
Ign:1 http://mirrors.esto.network/kali kali-rolling InRelease
Get:2 http://mirrors.esto.network/kali kali-rolling InRelease [34.0 kB]
Get:3 http://mirrors.esto.network/kali kali-rolling/main amd64 Packages [21.5 MB]
Ign:2 http://mirrors.esto.network/kali kali-rolling/main amd64 Packages
Get:4 http://mirrors.esto.network/kali kali-rolling/main amd64 Contents (deb) [53.8 MB]
Ign:3 http://mirrors.esto.network/kali kali-rolling/main amd64 Contents (deb)
Get:5 http://mirrors.esto.network/kali kali-rolling/contrib amd64 Packages [113 kB]
Get:6 http://mirrors.esto.network/kali kali-rolling/contrib amd64 Contents (deb) [261 kB]
Get:7 http://mirrors.esto.network/kali kali-rolling/non-free amd64 Packages [183 kB]
Get:8 http://mirrors.esto.network/kali kali-rolling/non-free amd64 Contents (deb) [915 kB]
Get:9 http://mirrors.esto.network/kali kali-rolling/non-free-firmware amd64 Packages [16.0 kB]
Get:10 http://mirrors.esto.network/kali kali-rolling/non-free-firmware amd64 Contents (deb) [40.6 kB]
Get:11 http://mirrors.esto.network/kali kali-rolling/main amd64 Packages [21.5 MB]
24% [2 Packages 3,553 kB/21.5 MB 17%] 2,853 PB/s 0s
```

## PRACTICAL TASKS & VERIFICATION (CONTINUED)

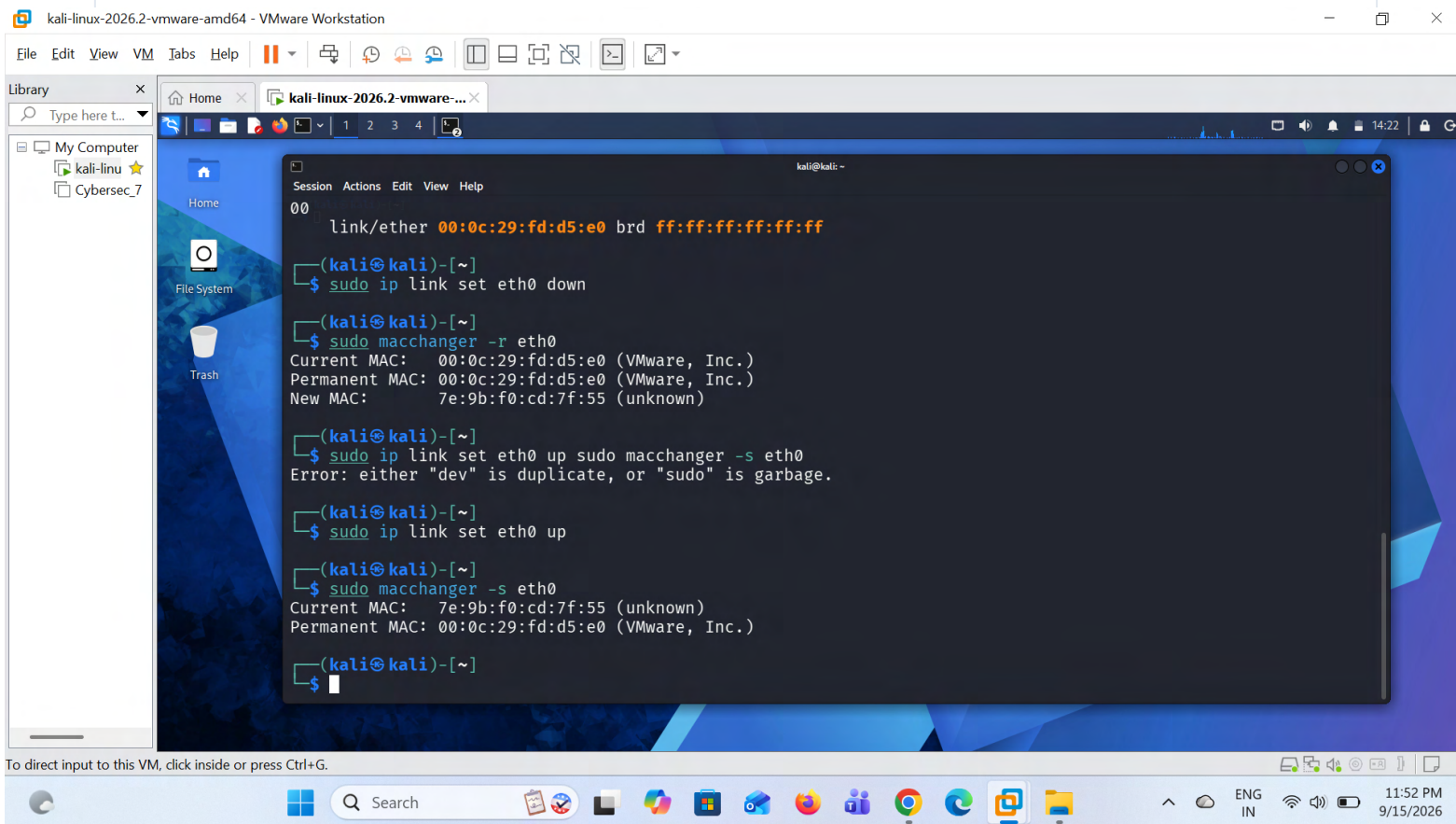
### Task 3: MAC Address Spoofing

**Cybersecurity Relevance:** A MAC address is a unique hardware identifier on layer 2. Penetration testers spoof MAC addresses to bypass MAC-filtering access controls, avoid forensic device tracking during wireless audits, and protect operational privacy.

#### Commands Executed:

```
$ sudo ip link set eth0 down
$ sudo macchanger -r eth0
$ sudo ip link set eth0 up
$ sudo macchanger -s eth0
```

**Observed Network Details:** Permanent MAC: 00:0c:29:fd:d5:e0 | Spoofed MAC: 7e:9b:f0:cd:7f:55



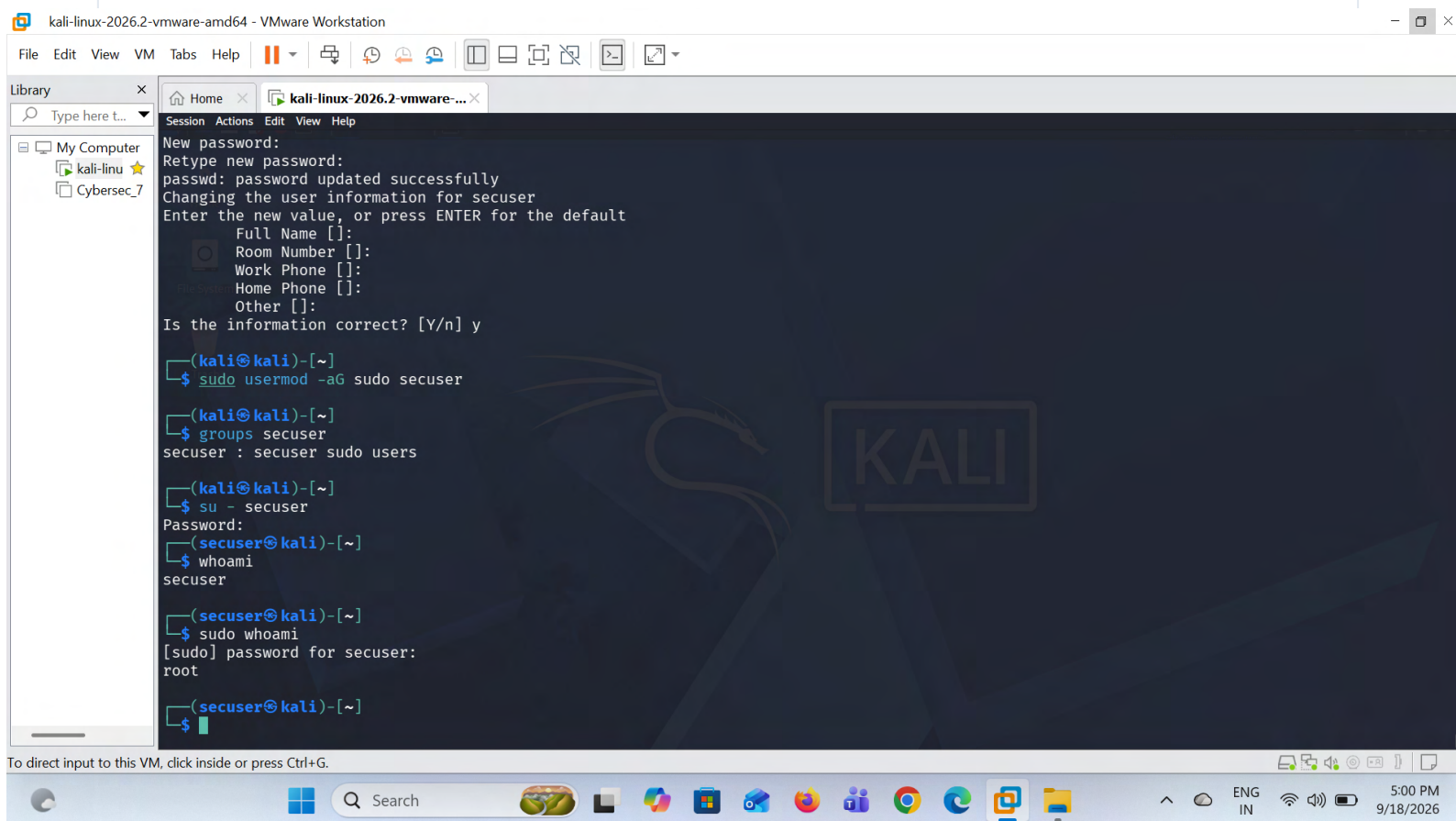
## PRACTICAL TASKS & VERIFICATION (CONTINUED)

### Task 4: User Management and Administrative Privileges

**Cybersecurity Relevance:** Adhering to the Principle of Least Privilege (PoLP) requires daily tasks to run under a standard user account rather than root. Delegating root access through the sudo group enables accountability and auditing in security operations.

#### Commands Executed:

```
$ sudo adduser secuser
$ sudo usermod -aG sudo secuser
$ groups secuser
$ su - secuser
$ whoami && sudo whoami
```



The screenshot shows a Kali Linux virtual machine running in VMware Workstation. The terminal window displays the following commands and output:

```
kali-linux-2026.2-vmware-amd64 - VMware Workstation
File Edit View VM Tabs Help
Library
My Computer
kali-linu
Cybersec.7
Session Actions Edit View Help
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for secuser
Enter the new value, or press ENTER for the default
Full Name []:
Room Number []:
Work Phone []:
Home Phone []:
Other []:
Is the information correct? [Y/n] y
(kali@kali)-[~]
$ sudo usermod -aG sudo secuser
(kali@kali)-[~]
$ groups secuser
secuser : secuser sudo users
(kali@kali)-[~]
$ su - secuser
Password:
(secuser@kali)-[~]
$ whoami
secuser
(secuser@kali)-[~]
$ sudo whoami
[sudo] password for secuser:
root
(secuser@kali)-[~]
$
```

The bottom of the screenshot shows the Windows taskbar with the Start button, Search bar, and various application icons. The system clock indicates 5:00 PM on 9/18/2026.

## 5. COMMANDS SUMMARY TABLE

| Task Category        | Primary Command                                    | Functional Purpose                                                           |
|----------------------|----------------------------------------------------|------------------------------------------------------------------------------|
| Identity & OS Info   | <code>whoami &amp;&amp; cat /etc/os-release</code> | Validates logged-in user context and verified Linux distribution release.    |
| Repository Update    | <code>sudo apt update</code>                       | Synchronizes local package index with the latest official Kali repositories. |
| MAC Manipulation     | <code>sudo macchanger -r eth0</code>               | Generates and assigns a randomized MAC address to interface eth0.            |
| User Provisioning    | <code>sudo adduser secuser</code>                  | Creates a new standard user account with designated home directory.          |
| Privilege Delegation | <code>sudo usermod -aG sudo secuser</code>         | Adds user to the sudoers group for administrative permission delegation.     |
| Session Switch       | <code>su - secuser</code>                          | Switches interactive login session to the newly created standard user.       |

## 6. REFLECTION & CONCLUSION

### What was Learned:

Through this minor project, practical hands-on expertise was gained in configuring an isolated penetration testing OS inside VMware Workstation. Essential administrative tasks—including keeping security toolchains updated, managing users according to the principle of least privilege, and manipulating Layer-2 network identifiers—were executed and validated successfully.

### Real-World Scenario Application:

- **Red Team Operations:** MAC spoofing provides operational anonymity during wireless access point auditing and physical security assessments.
- **Blue Team & Defense:** Regular system updating and strict user access controls help SOC analysts and system administrators prevent unauthorized privilege escalation.
- **Isolation & Testing:** Virtualization prevents live attack experiments from impacting the underlying host machine.